

- 4** The cubic equation $x^3 + 2x^2 + 3x + 3 = 0$ has roots α, β, γ .

(a) Find the value of $\alpha^2 + \beta^2 + \gamma^2$. [2]

[illegible]

(b) Show that $\alpha^3 + \beta^3 + \gamma^3 = 1$. [2]

[illegible]

- (c) Use standard results from the list of formulae (MF19) to show that

$$\sum_{r=1}^n ((\alpha+r)^3 + (\beta+r)^3 + (\gamma+r)^3) = n + \frac{1}{4}n(n+1)(an^2 + bn + c),$$

where a , b and c are constants to be determined.

[6]

[illegible]